Ch 7

Properties of Laplace eq'n

① $u \in C^0(\Omega)$ & $\Delta u = 0 \Rightarrow u \in C^\infty(\Omega) \Rightarrow u$ is harmonic

② <u>Mean-Value Property</u> (MVP)

If u is harmonic, then u is the average of the pts equal distance from it (if all those pts are in the domain)

$$u(\underline{r}_0) = \frac{1}{|S(\underline{r}_0, \delta)|} \int_{S(\underline{r}_0, \delta)} u(\underline{r})\, d\sigma$$

③ ~~Max~~ <u>Max. Principle</u>

Ω bdd in $\mathbb{R}^n$, $u \in C^0(\bar{\Omega})$, u harmonic.

Then u obtains its max & min. values on the bdry of Ω, $\partial\Omega$, and only on $\partial\Omega$ unless u is constant.

Ch 7 §6 Well-posedness of Dirichlet Problem

Dirichlet problem (for Laplace eq'n)

Let Ω be a regular domain (domain, bdry Lipschitz, bounded)

and $f \in C^0(\partial\Omega)$ be given

Find $u \in C^0(\bar{\Omega}) \cap C^2(\Omega)$ so that

$$\Delta u = 0 \quad \text{in } \Omega$$
$$u = f \quad \text{on } \partial\Omega$$

We need to show:

(i) sol'n is unique

(ii) sol'n exists

(iii) sol'n depends continuously on data (f)

(small changes in $f \Rightarrow$ small changes in u).

(i) To show: If u_1 and u_2 satisfy the Dirichlet problem, then $u_1 = u_2$. (shows uniqueness)

Proof. Let $\tilde{u} = u_1 - u_2$

Then $\tilde{u} \in C^0(\bar{\Omega}) \cap C^2(\Omega)$

$$\Delta \tilde{u} = \Delta(u_1 - u_2) = \Delta u_1 - \Delta u_2 = 0 - 0 = 0$$

laplace is a linear operator

u_1 & u_2 satisfy the D.P.

and on bdry, $\partial\Omega$ $\tilde{u} = u_1 - u_2 = f - f = 0$

↳ u_1 & u_2 satisfy the D.P.

By max. principle: Max & Min must occur on bdry

$\tilde{u} = 0$, on $\partial\Omega$. Max: 0, Min: 0

$\Rightarrow \tilde{u} = 0$ over $\bar{\Omega}$.

(ii) Let u_1 be the sol'n to D.P. with $f = f_1$

Let u_2 " " " " " " $f = f_2$

To show: If $\max_{x \in \partial\Omega} |f_1 - f_2| < \varepsilon$

Then $\max_{x \in \bar{\Omega}} |u_1 - u_2| < \varepsilon$

for using max principle

(i) Ω is bdd ✓

(ii) $\tilde{u} \in C^0(\bar{\Omega})$ ✓

(iii) $\Delta \tilde{u} = 0$ ✓

Let $\tilde{u} = u_1 - u_2$

Then $\tilde{u} \in C^0(\bar{\Omega}) \cap C^2(\Omega)$

$\Delta \tilde{u} = \Delta(u_1 - u_2) = \Delta u_1 - \Delta u_2$

linearity of Δ

$= 0 - 0 = 0$

u_1 & u_2 satisfy $\Delta u = 0$.

Max principle

Max + Min of $\tilde{u}$ must occur on the bdry.

on $\partial\Omega$ $\tilde{u} = f_1 - f_2$

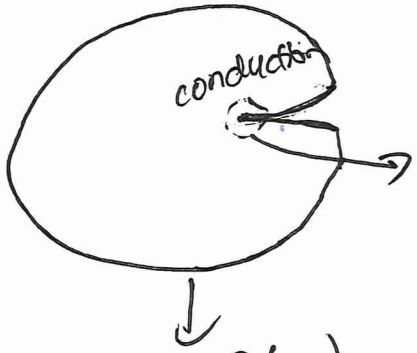
$$\left.\begin{array}{l} \max: |f_1 - f_2| < \varepsilon \\ \max: |\tilde{u}| < \varepsilon \end{array}\right\} \text{ on } \partial\Omega$$

$\Rightarrow |\tilde{u}| < \varepsilon$ on $\bar{\Omega}$.

(ii) Existence: More difficult to show, and depends on geometry of domain. (Not responsible for)

In fact, we do not have existence in some cases

2-D plate with crack

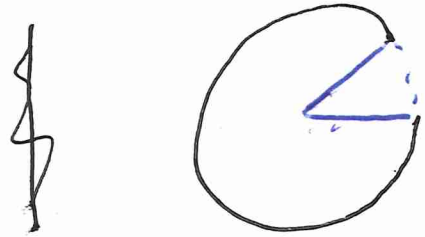


problem. not enough surface area.

$u = f(\underline{x}) \quad \underline{x} \in \partial\Omega$

To guarantee a sol'n, the crack's must satisfy "The cone condition"

probe condi'n



In <u>3-D</u> existence of the shape $y = x^k$ (rotate it)

<u>Existence</u>: If Ω is a bdd domain in $\mathbb{R}^n$ $(n=2,3)$ which satisfies the probe condition, then the D.P. has a sol'n!

$u_{xx} + u_{yy} - 5u_x + ye^x u_y - xyu = 0$

Ch 7 §8 Fourier Series

(we will cover Section 7 next $\Rightarrow$ solving $\Delta u = 0$ ~~on a~~ in polar coord on a finite domain).

Recall from Calc 2, we can represent certain fns in terms of Taylor series (as an ∞-long polynomial)

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2!} f''(x_0)(x - x_0)^2 + \cdots$$

ex) $\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$ $\quad (x_0 = 0)$

Q: What kind of fns have a Taylor series $\Rightarrow C^\infty(I)$, $x_0 \in I$

Q: Does the Taylor series converge to $f(x)$?

Not all C^∞ f'ns do: (not responsible for)

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$f(x) \in C^\infty$ (even